TimelyGPT Architecture

1. Using TimelyGPT as the generation and prediction module to replace TrajGPT
2. Compared to the original plan, the number of attention heads is increased to 8, and the model dimension is raised to 64
3. Using Time Embedding instead of RoPE: Assign a learnable time embedding vector to each time point and add it to the Cell Embedding

TimelyGPT Architecture

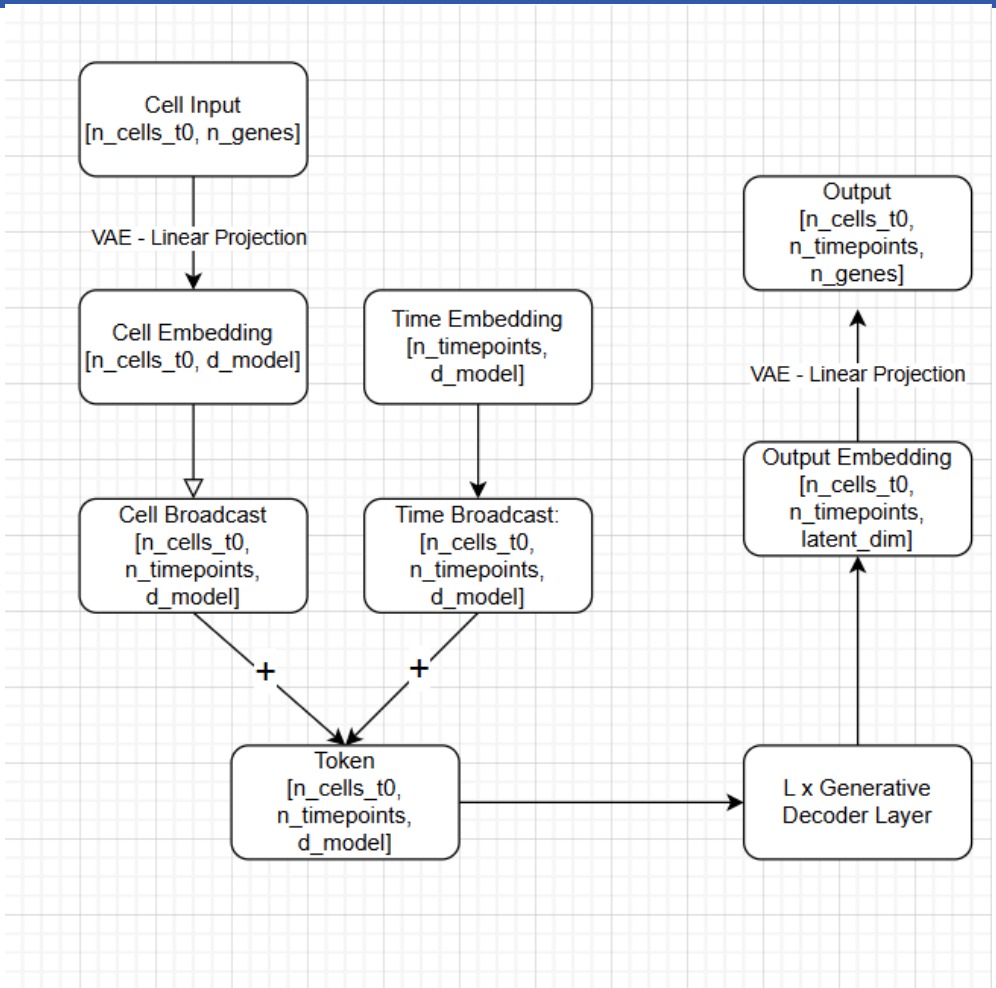


Figure 1: TimelyGPT Applied to SingleCell Data

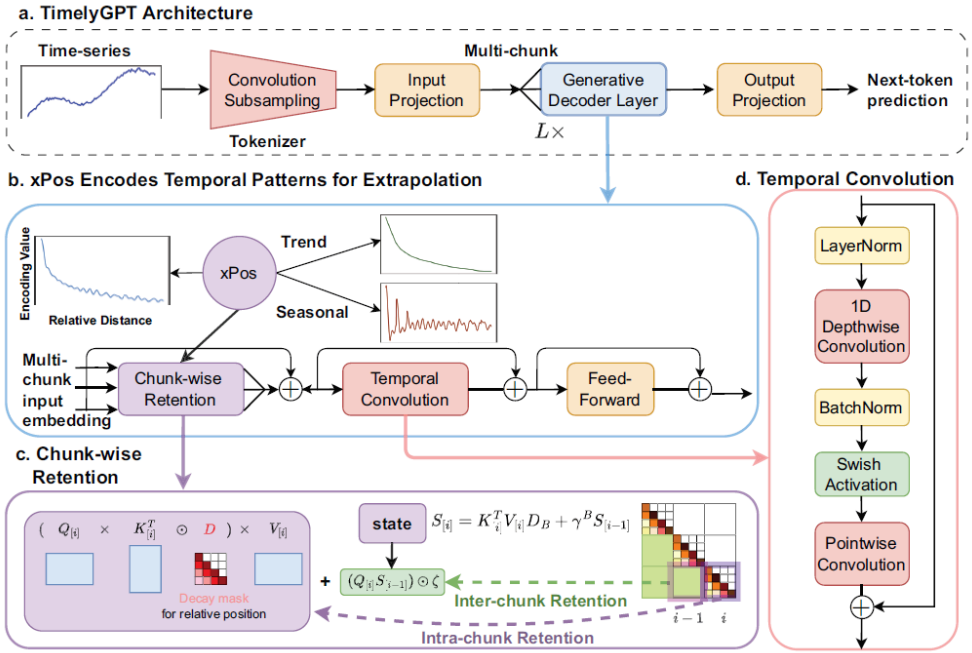


Figure 2: Original TimelyGPT Architecture

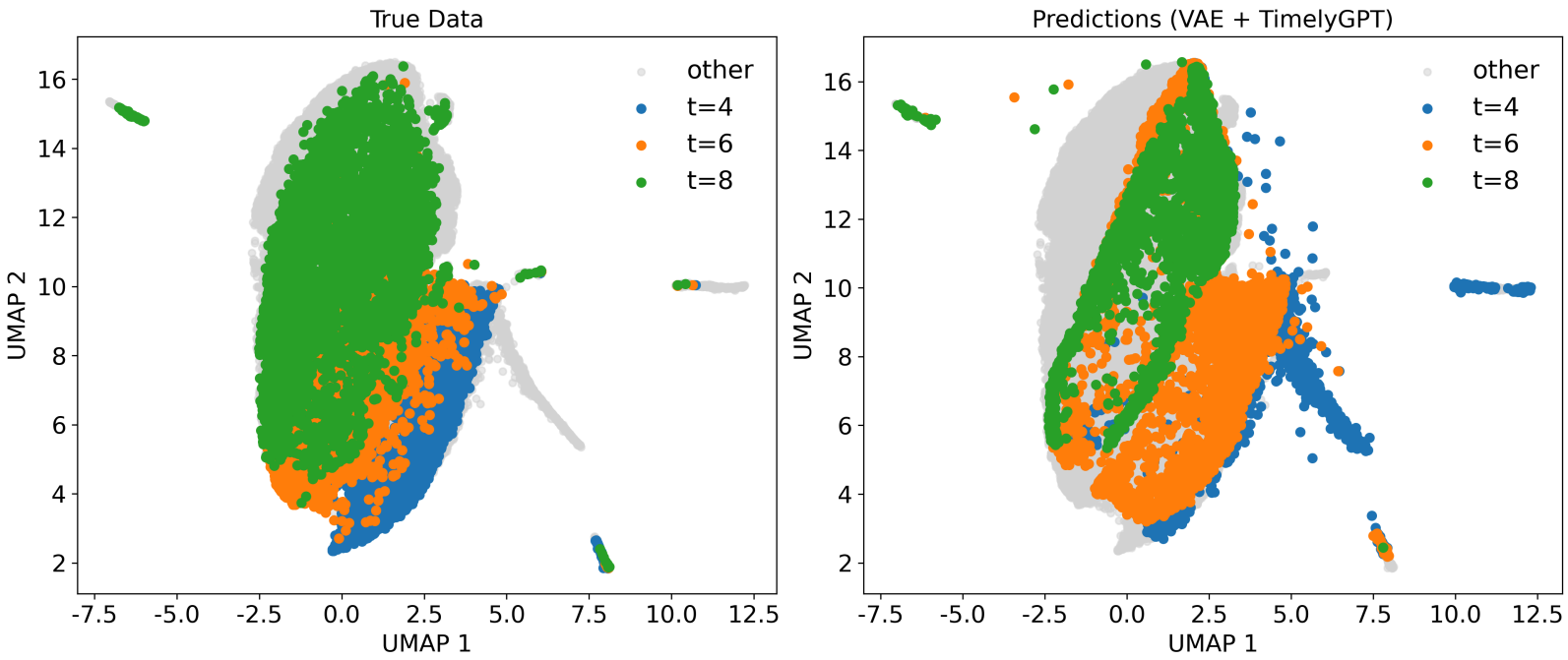
TimelyGPT Prediction

In the current solution, I use a prediction algorithm similar to scNODE:

1. Use VAE to generate a batch of 5000 cells at time t0 that follow the same distribution as the cells in the data at t0
2. Copy this batch of cells to every time point, add time embedding to create differentiation, thus generating Tokens
3. Input Tokens into the TimelyGPT prediction module, output predicted cell embeddings for each time point
4. VAE decodes the cell embeddings, calculate loss using optimal transport (Wasserstein distance, Sinkhorn algorithm)

This algorithm is called ‘similar to scNODE’ because it hasn’t escaped the Neural ODE mindset of scNODE: starting from an initial state z0 at time t0, it deterministically generates the subsequent sequence, where the initial state z0 determines the entire sequence

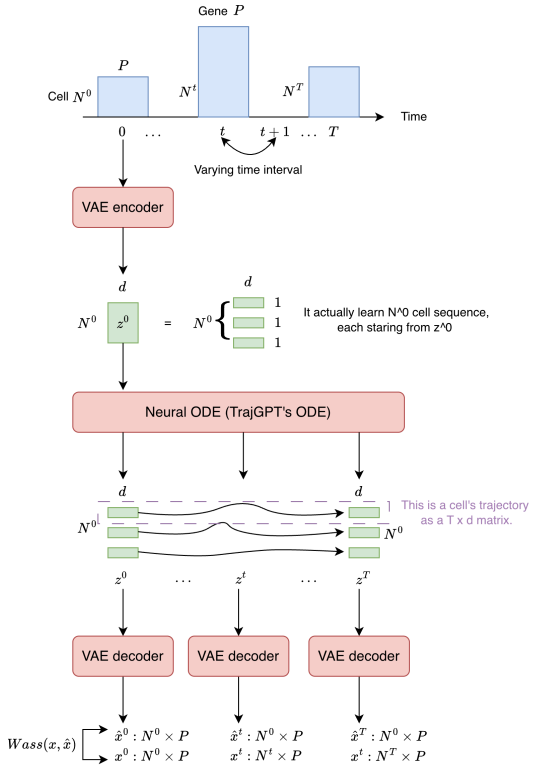
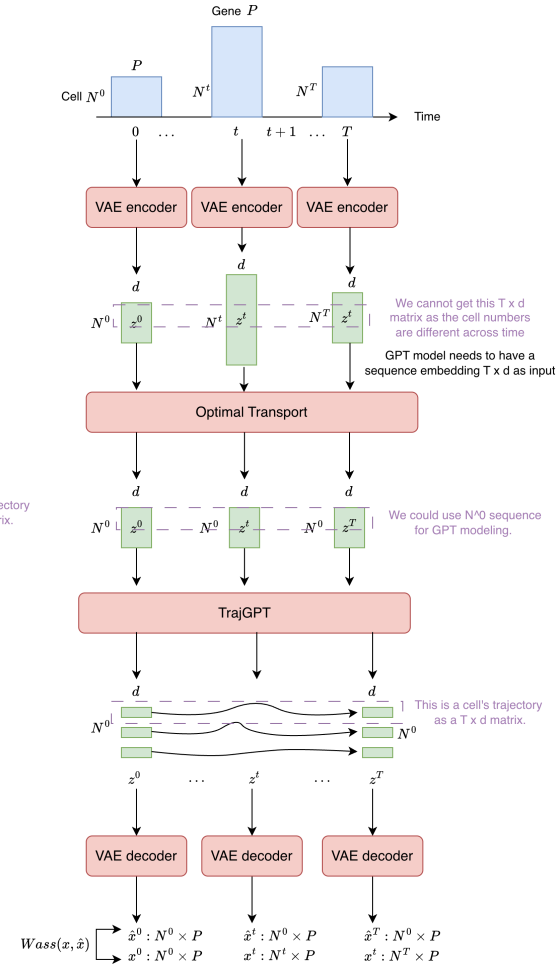
Current TimelyGPT Prediction Results



Comment: The results are quite good and can be further optimized, but training is very slow, and the computer can't handle it

Optimization Plan and Current Problems

1. After discussion, we believe the first optimization point is to use a GPT-style generation method: input tokens t0,...,tn-1, output token tn, and iterate
2. Since it's no longer necessary to broadcast z0 to all time points, we can simultaneously replace the original time embedding with RoPE or xPOS encoding

- The implementation problem with the above solution is that GPT requires a (uniform) sequence, but the cell number at each time point is different. This might require an additional Optimal Transport Alignment

Optimization Plan and Current Problems